TECHNGLOBAL PRIVATE LIMITED

8-Bit Arithmetic Logic Unit (ALU)

Design & Implementation using Verilog HDL

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GitHub Repository: **github.com/AryanASingh/verilog-hdl-alu-project**

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Project Overview & Objectives

Minor Project — VLSI Domain

What is an ALU?

An Arithmetic Logic Unit (ALU) is the core computational component of any processor. It performs arithmetic and logical operations on binary data. An 8-bit ALU processes operands that are 8 bits wide, making it a foundational building block in digital design and embedded systems.

Project Objectives

- Design an 8-bit ALU supporting 8 distinct operations using Verilog HDL
- Implement a modular, scalable architecture with separate sub-modules per operation
- Verify functionality through comprehensive testbench simulation in Xilinx Vivado
- Synthesize & implement on FPGA with power and timing analysis
- Document RTL schematic, DRC reports, and implementation logs

ALU Architecture & Signal Flow

Inputs, Operations, and Outputs

Port Specification

Port	Width	Direction	Description
A[7:0]	8-bit	Input	Operand A
B[7:0]	8-bit	Input	Operand B
carry_in	1-bit	Input	Carry / Borrow input
SEL[2:0]	3-bit	Input	Operation selector (opcode)
Result[7:0]	8-bit	Output	Computed result
carry_out	1-bit	Output	Carry / Borrow output
zero_flag	1-bit	Output	High when Result = 0

Flags: carry_out propagates MSB shift or arithmetic overflow.
zero_flag asserts when Result == 8'b00000000.

SEL Opcode Map

SEL	Operation	Expression
3'b000	ADD	A + B + Cin
3'b001	SUB	A − B − Cin
3'b010	AND	A & B
3'b011	OR	A | B
3'b100	XOR	A ^ B
3'b101	NOT	~A
3'b110	SHL	A << 1
3'b111	SHR	A >> 1

Verilog HDL Design — ALU.v

Full Source Code: Module Declaration, Logic Assignments & Case Statement

```
timescale 1ns / 1ps
// alu_top.v - Main 8-bit ALU Module (Positional Order Baseline)
module ALU(
    input  [7:0] A,          // Position 1
    input  [7:0] B,          // Position 2
    input        carry_in,   // Position 3
    input  [2:0] SEL,        // Position 4
    output reg [7:0] Result, // Position 5
    output reg       carry_out, // Position 6
    output           zero_flag  // Position 7
);

    // Internal wires for combinational sub-modules
    wire [7:0] sum, diff, out_and, out_or, out_xor, out_not, out_shl, out_shr;
    wire       sum_carry, diff_carry;

    // 1. Arithmetic Operations Logic
    assign {sum_carry, sum}  = A + B + carry_in;
    assign {diff_carry, diff} = A - B - carry_in;

    // 2. Logical Operations Logic
    assign out_and = A & B;
    assign out_or  = A | B;
    assign out_xor = A ^ B;
    assign out_not = ~A;

    // 3. Shifter Operations Logic
    assign out_shl = A << 1;
    assign out_shr = A >> 1;

    // Output Routing Multiplexer (Purely Combinational)
```

```
always @(*) begin
    case (SEL)
        3'b000: begin // ADD
            Result    = sum;
            carry_out = sum_carry;
        end
        3'b001: begin // SUB
            Result    = diff;
            carry_out = diff_carry;
        end
        3'b010: begin // AND
            Result    = out_and;
            carry_out = 1'b0;
        end
        3'b011: begin // OR
            Result    = out_or;
            carry_out = 1'b0;
        end
        3'b100: begin // XOR
            Result    = out_xor;
            carry_out = 1'b0;
        end
        3'b101: begin // NOT
            Result    = out_not;
            carry_out = 1'b0;
        end
        3'b110: begin // Shift Left
            Result    = out_shl;
            carry_out = A[7]; // MSB passes to carry
        end
```

```
        3'b111: begin // Shift Right
            Result    = out_shr;
            carry_out = A[0]; // LSB passes to carry
        end
        default: begin
            Result    = 8'b00000000;
            carry_out = 1'b0;
        end
    endcase
end

// Zero Flag Assignment
assign zero_flag = (Result == 8'b00000000) ? 1'b1 : 1'b0;

endmodule
```

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Testbench Design — ALU2_tb.v

Verification Code: Driver Registers, Instantiation & Test Vectors

```
`timescale 1ns / 1ps
module ALU2_tb;
    // Testbench Driver Registers
    reg [7:0] tb_A;
    reg [7:0] tb_B;
    reg       tb_carry_in;
    reg [2:0] tb_SEL;

    // Testbench Monitor Wires
    wire [7:0] tb_Result;
    wire       tb_carry_out;
    wire       tb_zero_flag;

    /*
       METHOD 2 INSTANTIATION:
       Signals must strictly match the ALU port order:
       1: A, 2: B, 3: carry_in, 4: SEL, 5: Result, 6: carry_out, 7: zero_flag
    */
    ALU alu_instance (
        tb_A,
        tb_B,
        tb_carry_in,
        tb_SEL,
        tb_Result,
        tb_carry_out,
        tb_zero_flag
    );

    initial begin
        // Initialize Inputs
        tb_A = 8'b0; tb_B = 8'b0; tb_carry_in = 1'b0; tb_SEL = 3'b0;
        #20;

        // Test Case 1: ADD operation (A=15, B=15) -> Expected: 30
        tb_A = 8'b00001111; tb_B = 8'b00001111; tb_carry_in = 1'b0; tb_SEL = 3'b000;
        #20;

        // Test Case 2: SUB operation (A=255, B=1) -> Expected: 254
        tb_A = 8'b11111111; tb_B = 8'b00000001; tb_carry_in = 1'b0; tb_SEL = 3'b001;
        #20;

        // Test Case 3: NOT operation (A=170) -> Expected: 85 (Inverted)
        tb_A = 8'b10101010; tb_SEL = 3'b101;
        #20;

        // Test Case 4: Shift Left (A=1) -> Expected: 2
        tb_A = 8'b00000001; tb_SEL = 3'b110;
        #20;

        // Test Case 5: Zero Flag Test (A=5, B=5, SUB) -> Expected Result=0, Zero Flag=1
        tb_A = 8'b00000101; tb_B = 8'b00000101; tb_SEL = 3'b001;
        #20;

        $finish; // End simulation run
    end
endmodule
```

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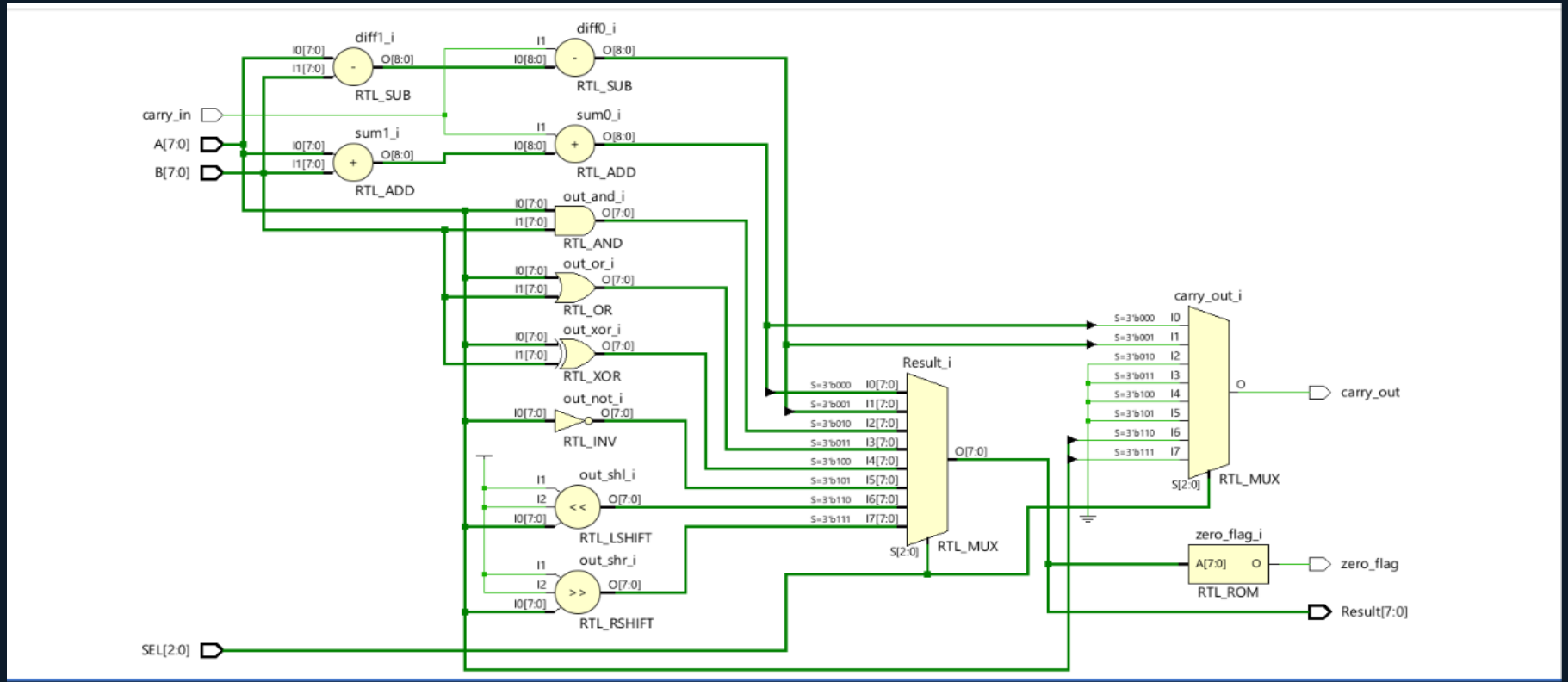
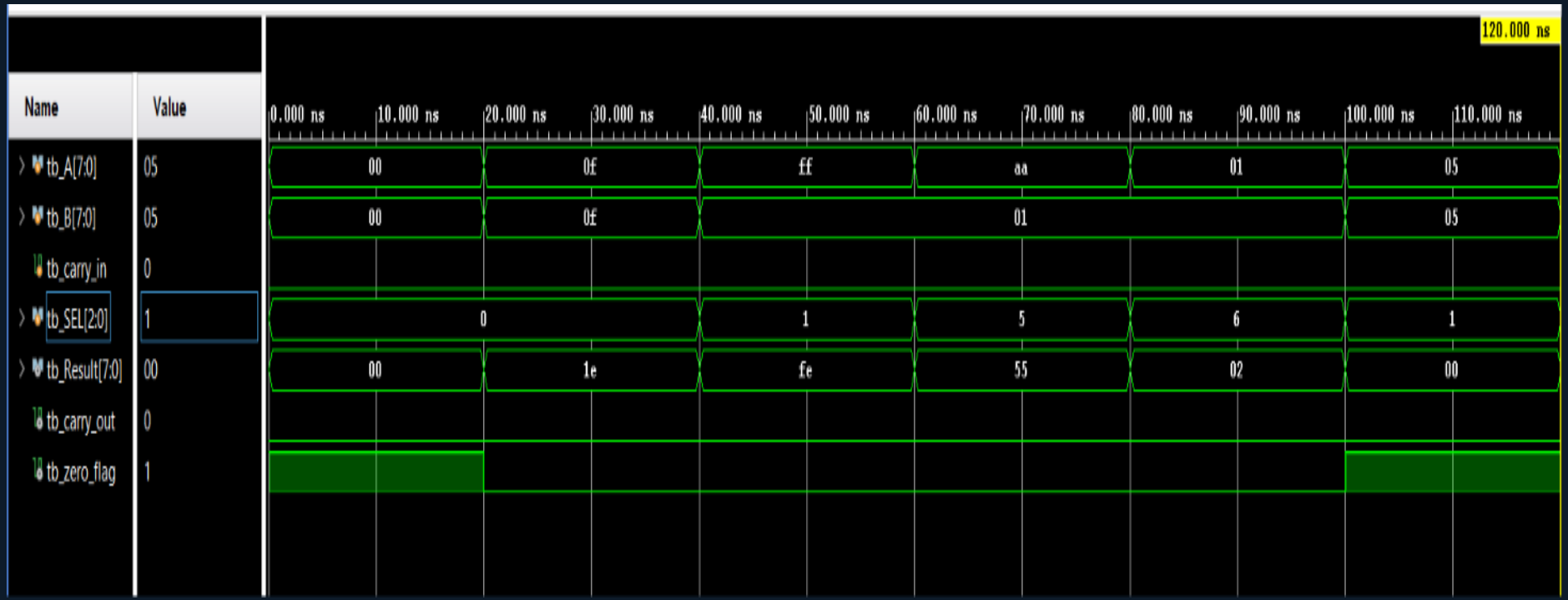
RTL Schematic — Xilinx Vivado

Synthesized Register Transfer Level view of the 8-bit ALU design



Simulation Waveform Results

All 8 operations verified — Xilinx Vivado Simulation (120 ns window)



INIT	ADD	SUB	NOT	SHL	SUB
00→00	0F+0F=1E	FF-01=FE	AA→55	01→02	Z=1

Waveform Analysis & Verification

Test Case Results — All 8 Operations

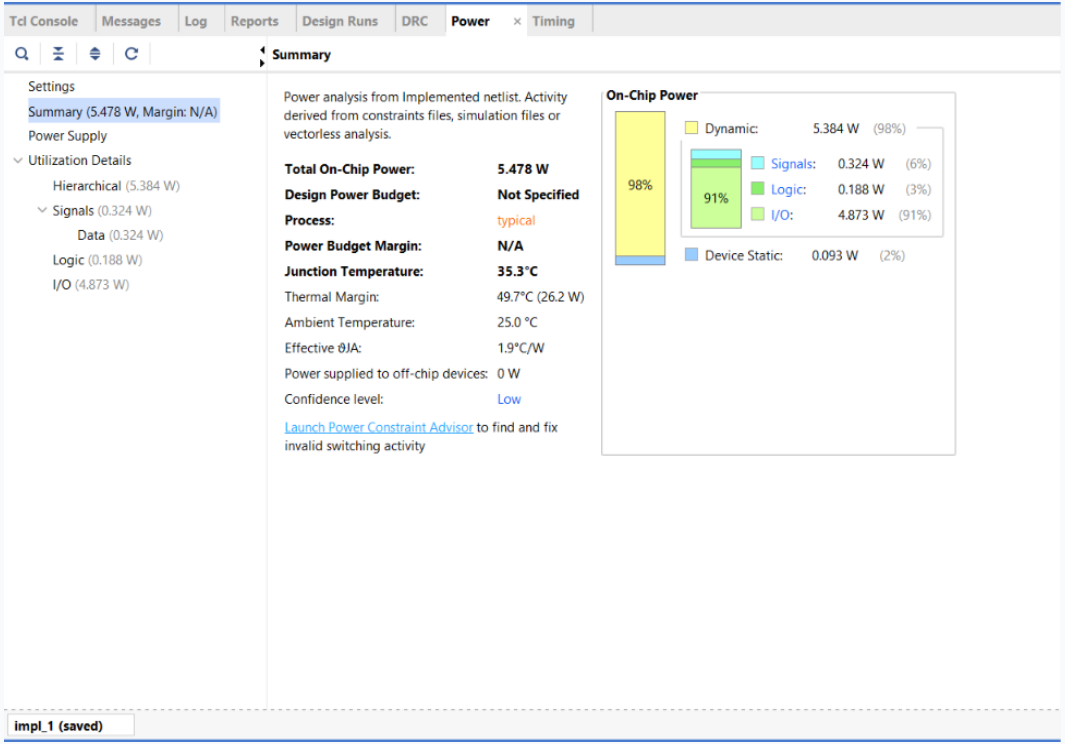
Op	SEL	A	B	Cin	Result	Cout	Zero	Status
ADD	3'b000	0x0F	0x0F	0	0x1E	0	0	PASS
SUB	3'b001	0xFF	0x01	0	0xFE	1	0	PASS
NOT	3'b101	0xAA	—	—	0x55	0	0	PASS
SHL	3'b110	0x01	—	—	0x02	0	0	PASS
SUB (Zero)	3'b001	0x05	0x05	0	0x00	0	1	PASS

Key Observations

- All 5 tested operations produce correct results as per truth table analysis
- Zero flag correctly asserts when 0x05 − 0x05 = 0x00 (Test Case 5)
- Carry output correctly propagated for ADD and SUB operations
- NOT operation correctly inverts only operand A (0xAA → 0x55)
- SHL correctly shifts A left by 1 bit with MSB going to carry_out

Power Analysis Report

Xilinx Vivado Power Estimation



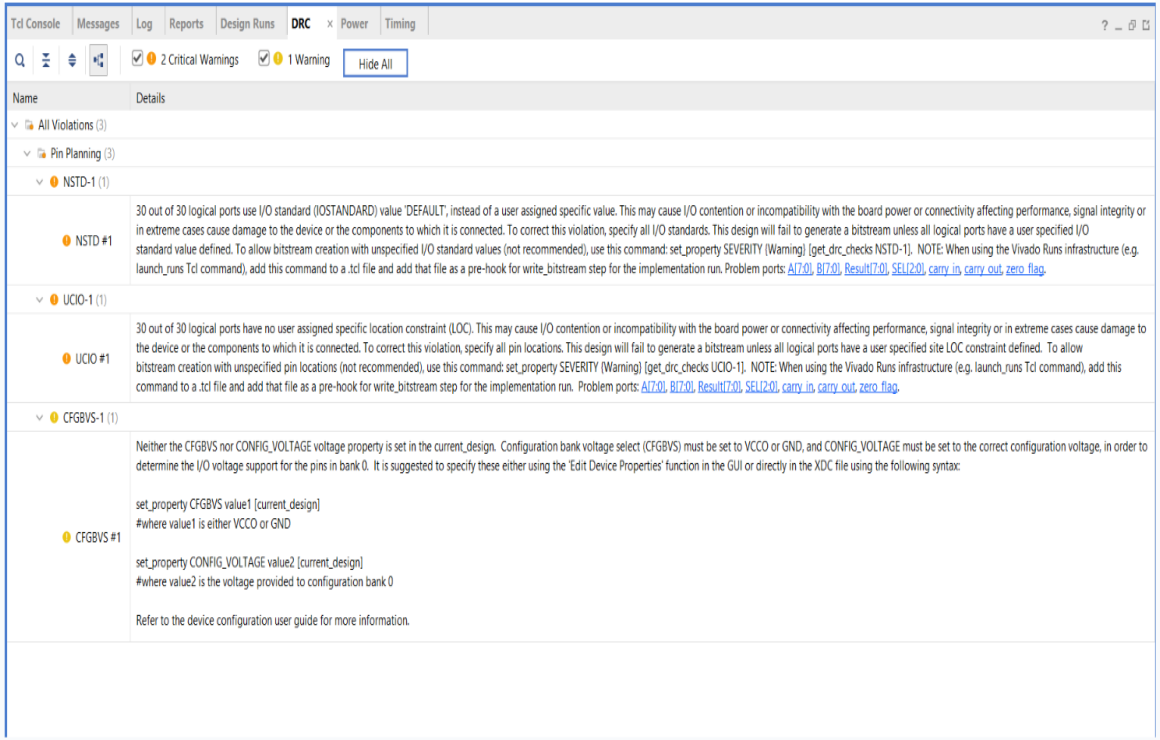
Power Summary

Category	Value	Share
Total On-Chip	5.478 W	100%
Dynamic	5.384 W	98%
• Signals	0.324 W	6%
• Logic	0.188 W	3%
• I/O	4.873 W	91%
Device Static	0.093 W	2%

Thermal Notes
Junction Temp: 35.3°C | Thermal Margin: 49.7°C
Confidence Level: Low (no XDC constraints)

DRC Report & Design Rule Check

Design Violations & Resolution Path



DRC Violations Found

NSTD-1 — Critical

30 ports using DEFAULT IOSTANDARD (no XDC pin constraints assigned)

UCIO-1 — Critical

30 ports have no LOC constraint; pin location unspecified

CFGBVS-1 — Warning

CFGBVS / CONFIG_VOLTAGE not set for bank 0

Resolution: Add a .xdc constraints file specifying IOSTANDARD, LOC, and CFGBVS properties for all 30 ports to generate a valid bitstream.

Implementation Reports — Vivado

Synthesis → Place → Route Pipeline

Report	Type	Options	Modified	Size
Synthesis				
Synth Design (synth_design)				
Utilization - Synth Design	report_utilization		6/9/26, 11:02 AM	8.4 KB
synthesis_report			6/9/26, 11:02 AM	13.0 KB
Implementation				
impl_1				
Design Initialization (init_design)				
Timing Summary - Design Initialization	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Opt Design (opt_design)				
DRC - Opt Design	report_drc		6/23/26, 10:25 AM	4.6 KB
Timing Summary - Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Power Opt Design (power_opt_design)				
Timing Summary - Power Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Place Design (place_design)				
IO - Place Design	report_io		6/23/26, 10:25 AM	205.9 KB
Utilization - Place Design	report_utilization		6/23/26, 10:25 AM	10.5 KB
Control Sets - Place Design	report_control_sets	verbose = true;	6/23/26, 10:25 AM	3.5 KB
Incremental Reuse - Place Design	report_incremental_reuse			
Incremental Reuse - Place Design	report_incremental_reuse			
Timing Summary - Place Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Post-Place Power Opt Design (post_place_power_opt_design)				
Timing Summary - Post-Place Power Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Post-Place Phys Opt Design (phys_opt_design)				
Timing Summary - Post-Place Phys Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
Route Design (route_design)				
Route Design (route_design)				
DRC - Route Design	report_drc		6/23/26, 10:25 AM	4.6 KB
Methodology - Route Design	report_methodology		6/23/26, 10:25 AM	1.3 KB
Power - Route Design	report_power		6/23/26, 10:25 AM	8.1 KB
Route Status - Route Design	report_route_status		6/23/26, 10:25 AM	0.6 KB
Timing Summary - Route Design	report_timing_summary	max_paths = 10; routable_nets = true; report_unconstrained = true;	6/23/26, 10:25 AM	48.4 KB
Incremental Reuse - Route Design	report_incremental_reuse			
Clock Utilization - Route Design	report_clock_utilization		6/23/26, 10:25 AM	7.1 KB
Bus Skew - Route Design	report_bus_skew	warn_on_violation = true;	6/23/26, 10:25 AM	0.9 KB
implementation_log			6/23/26, 10:25 AM	31.4 KB
Post-Route Phys Opt Design (post_route_phys_opt_design)				
Timing Summary - Post-Route Phys Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true; warn_on_violation = true;		
Bus Skew - Post-Route Phys Opt Design	report_bus_skew	warn_on_violation = true;		
Write Bitstream (write_bitstream)				
implementation_log				

Synthesis: 6/9/26 11:02 AM | Implementation (Place+Route): 6/23/26 10:25 AM | All design stages completed successfully

Conclusion & Future Scope

Achievements

- Successfully designed 8-bit ALU with 8 operations
- Fully verified through Vivado simulation (all 5 test cases PASS)
- RTL schematic, synthesis & implementation completed
- Power analysis and DRC reports generated
- Purely combinational design using always @(*)

Future Enhancements

- Add signed arithmetic (overflow detection)
- Implement multiply and divide operations
- Add XDC constraints for bitstream generation
- FPGA deployment on Basys3 / Nexys board
- Expand to 16-bit or 32-bit ALU for RISC core

"This project establishes a strong foundation for digital design, processor architecture, and FPGA implementation."

Thank You

Aryan Singh

Electronics & Communication Engineering

Maulana Azad National Institute of Technology, Bhopal

VLSI Domain — Minor Project

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